

Lengths and Ratios

Diana Tolu

June, 2025

Definition

Every point in the plane corresponds to a complex number $z = a + bi$, where:

- $a = \operatorname{Re}(z)$ is the real part
- $b = \operatorname{Im}(z)$ is the imaginary part
- $|z| = \sqrt{a^2 + b^2}$ is the magnitude
- $\bar{z} = a - bi$ is the conjugate

General formulas

Perpendicularity: Let A, B, C, D be pairwise distinct points. Then $AB \perp CD$ if and only if

$$\frac{d-c}{b-a} \in i\mathbb{R} \quad \text{or equivalently} \quad \frac{d-c}{b-a} + \overline{\left(\frac{d-c}{b-a}\right)} = 0.$$

Collinearity: Let A, B, C be pairwise distinct points. Then A, B, C are collinear if and only if

$$\frac{c-a}{c-b} \in \mathbb{R} \quad \text{or equivalently} \quad \frac{c-a}{c-b} = \overline{\left(\frac{c-a}{c-b}\right)}.$$

Concyclicity: Let A, B, C, D be pairwise distinct points. Then A, B, C, D are concyclic if and only if

$$\frac{c-a}{c-b} : \frac{d-a}{d-b} \in \mathbb{R}.$$

Reflection About a Segment: Let W be the reflection of Z across the segment AB . Then

$$w = \frac{(a-b)\bar{z} + \bar{a}b - a\bar{b}}{\bar{a} - \bar{b}}.$$

The foot of the perpendicular from Z to line AB is

$$\frac{1}{2}(w + z).$$

Complex Shoelace Formula: Let A, B, C be points in the complex plane. The signed area of triangle $\triangle ABC$ is

$$\text{Area} = \frac{i}{4} \begin{vmatrix} a & \bar{a} & 1 \\ b & \bar{b} & 1 \\ c & \bar{c} & 1 \end{vmatrix}.$$

In particular, A, B, C are collinear if and only if this determinant is zero.

Intersection of Lines: Let A, B, C, D be points. Then the intersection point of lines AB and CD is

$$\frac{(\bar{a}b - a\bar{b})(c-d) - (a-b)(\bar{c}d - c\bar{d})}{(\bar{a} - \bar{b})(c-d) - (a-b)(\bar{c} - \bar{d})}.$$

The Unit Circle and Triangle Centers

The unit circle in the complex plane is defined by $|z| = 1$. A key identity for such points is:

$$z = \frac{1}{\bar{z}} \quad \text{when } |z| = 1.$$

Triangle Centers (Assume ABC lies on the unit circle)

- Circumcenter: $O = 0$
- Centroid: $G = \frac{1}{3}(a + b + c)$
- Orthocenter: $H = a + b + c$

These points lie on the Euler line.

Reflection Over Chord AB (with $|a| = |b| = 1$)

- Reflection of point z : $a + b - ab\bar{z}$
- Foot from z to line AB : $\frac{1}{2}(z + a + b - ab\bar{z})$

Intersection of Chords AB and CD on Unit Circle

$$\frac{ab(c + d) - cd(a + b)}{ab - cd}$$

Incenter of Triangle

Let $a = x^2$, $b = y^2$, $c = z^2$ with $|x| = |y| = |z| = 1$. Then the incenter I is

$$I = -(xy + yz + zx).$$

Problem 1

Let A, B lie on the unit circle and let P be a point such that PA and PB are tangents to the circle. Then

$$P = \frac{2ab}{a + b}.$$

Problem 2

A quadrilateral $ABCD$ is inscribed in the unit circle. If $P = AB \cap CD$ and $Q = AD \cap BC$, then prove that

$$p \cdot \bar{q} + q \cdot \bar{p} = 2.$$

Problem 3

Let H be the orthocenter of triangle ABC . Let D, E, F lie on the circumcircle of ABC such that $AD \parallel BE \parallel CF$. Let S, T, U respectively denote the reflections of D, E, F across BC, CA, AB . Prove that the points S, T, U, H are concyclic.

Problem 4 (Steiner Line)

Let ABC be a triangle and let P be a point on its circumcircle. Let X, Y, Z be the reflections of P over the lines BC, CA, AB , respectively. Prove that the points X, Y, Z, H are collinear, where H is the orthocenter of triangle ABC .

Problem 5

Let $ABCD$ be a cyclic quadrilateral. Let H_A, H_B, H_C, H_D denote the orthocenters of triangles $\triangle BCD, \triangle CDA, \triangle DAB, \triangle ABC$, respectively. Show that the lines AH_A, BH_B, CH_C, DH_D are concurrent.

Problem 6 (Brokard)

Let $ABCD$ be a cyclic quadrilateral with center O . Let $E = AC \cap BD$, $F = AB \cap CD$, and $G = AD \cap BC$. Prove that O is the orthocenter of triangle EFG .

Problem 7

In triangle ABC with incenter I , the incircle is tangent to CA and AB at points E and F , respectively. Let G and H be the reflections of E and F across I . Let $Q = BC \cap GH$, and let M be the midpoint of segment BC . Prove that $IQ \perp IM$.

Problem 8

Let O be the circumcenter of triangle ABC . A line ℓ through O intersects AB and AC at points X and Y , respectively. Let M and N be the midpoints of BY and CX , respectively. Show that $\angle MON = \angle BAC$.

Problem 9

Let $\triangle ABC$ be acute, and let M be the midpoint of the minor arc $\widehat{BC}$. Let N be a point on the circumcircle of ABC such that $AN \perp BC$, and let K, L lie on AB, AC such that $OK \parallel MB$ and $OL \parallel MC$, where O is the circumcenter of ABC . Prove that $NK = NL$.

Problem 10

Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment PQ . Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.